AP PRECALCULUS

Unit 1 • Worksheet 1.9-1.10B • Answer key

Worksheet 1.9-1.10B • Rational Functions and Asymptotes

Answer key with worked solutions

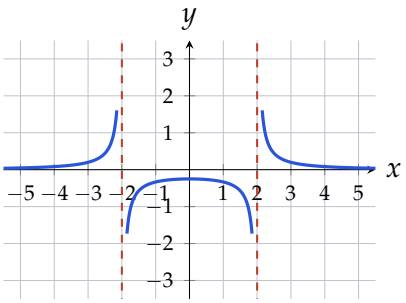
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The graph of $r(x) = \dfrac{1}{x^2 - 4}$ is shown. Which statement is true?



(A) $\lim_{x \to 2^-} r(x) = \infty$

(B) $\lim_{x \to 2^+} r(x) = -\infty$

(C) $\lim_{x \to -2^+} r(x) = \infty$

(D) $\lim_{x \to 2^-} r(x) = -\infty$

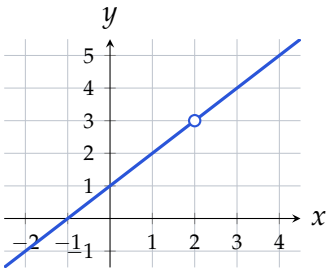
Solution

Answer: (D). Between the asymptotes the graph falls without bound near $x = 2$.

Question 2

NO CALCULATOR

The graph of a rational function is a line with a hole at $(2,3)$. Which could be its equation?



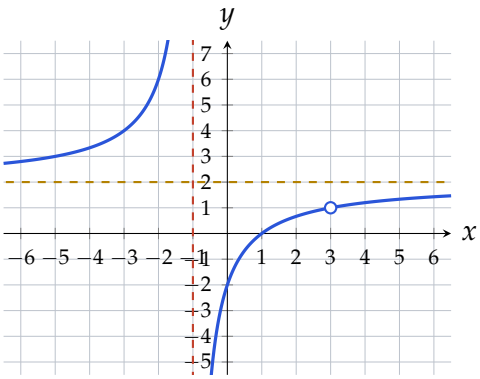
(A) $\dfrac{x^2-4}{x-2}$
(B) $\dfrac{x+1}{x-2}$
(C) $\dfrac{x^2-x-2}{x-2}$
(D) $x+1$

Solution
Answer: (C). $\frac{(x-2)(x+1)}{x-2}$.

Question 3 (Free response)

NO CALCULATOR

The graph of r is shown. It has a vertical asymptote $x=-1$, a horizontal asymptote $y=2$, a zero at $x=1$ and a hole at $(3,1)$.



(a) Write limit statements describing r near $x=-1$, at $x=3$, and as $x\to\pm\infty$.
(b) Write an expression for $r(x)$ that has all of these features.
(c) Solve $r(x)<0$.

Solution
(a) $\lim_{x\to -1^-} r = \infty$, $\lim_{x\to -1^+} r = -\infty$; $\lim_{x\to 3} r = 1$; $\lim_{x\to\pm\infty} r = 2$.
(b) $r(x) = \dfrac{2(x-1)(x-3)}{(x+1)(x-3)}$ (check: $\frac{2(2)}{4} = 1$ at $x=3$).
(c) $(-1,1)$.